

Approximate Unlearning: Mitigating Social Bias in SLMs and SSMs through Comparative Evaluation

Team Name: BERT's Buddies

Snavya Prasad, Nicolas Burton, Ashmit Pai, Nikhil Kanodia, Remi Selvam, Sanskar Srivastava

Mentor: Fenil Bardoliya

CSE 576: Topics in NLP

Abstract

This research addresses the challenge of social bias in language models by evaluating approximate machine unlearning techniques as a mitigation strategy. We focus on Small Language Models (SLMs) and State Space Models (SSMs), testing whether targeted parameter updates can reduce harmful biases while preserving model utility. Our study compares five unlearning methods: Task Vector Negation, Partitioned Contrastive Gradient Unlearning, Gradient Ascent, Low Rank Adaptation, and Selective Knowledge Unlearning, across datasets including BBQ, CrowS Pairs, StereoSet, WinoGender, PKU-SafeRLHF, LLM LAT, and TOFU. Experiments conducted on Phi-3.5 mini instruct (SLM), Qwen2.5 0.5B (transformer baseline), and Mamba 130M (SSM) demonstrate substantial reductions in bias scores across multiple demographic categories, with some techniques achieving over 40% improvement. However, we also observe trade-offs in fluency and content fidelity depending on the method. This work presents a comparative analysis of parameter efficient unlearning strategies and highlights their promise for building safer and fairer language models without full retraining.

1 Introduction

1.1 Problem Statement:

Natural language processing (NLP) models, including Small Language Models (SLMs) and State-Space Models (SSMs), are increasingly used in decision-making and communication tasks. However, these models often inherit and amplify social biases—such as those based on race, gender, and socioeconomic status—present in their training data. This can result in unfair, exclusionary, or even harmful outcomes, especially when deployed in sensitive real-world applications like hiring, lending, or content moderation.

Why Bias Is Inherent in Language Models? Bias in language models arises primarily from two sources:

Intrinsic Bias: This is embedded during model training, as models learn from large datasets that reflect societal prejudices and imbalances. For example, models may associate certain professions with specific genders or underrepresent minority groups, leading to skewed outputs.

Extrinsic Bias: This manifests during real-world use, where models may produce biased responses or decisions in downstream tasks, such as disproportionately flagging certain dialects as offensive or reinforcing stereotypes in sentiment analysis.

These biases can negatively impact individuals by perpetuating stereotypes, marginalizing groups, and producing unfair outcomes. For models, unchecked bias can degrade trust, reduce fairness, and compromise performance, particularly on tasks involving diverse populations.

Ideally, NLP models should operate fairly and equitably across all demographic groups, producing outputs that are free from harmful social biases. To address this, our research explores the use of approximate machine unlearning techniques that can selectively remove biased behaviors from trained models while preserving their overall language understanding and utility.

1.2 Prior Work

Recent research in bias mitigation for language models has introduced a range of post hoc unlearning and parameter editing techniques aimed at reducing social biases without requiring full model retraining. These approaches, combined with specialized datasets and evaluation tools, have advanced the field by improving computational efficiency and allowing more targeted interventions. However, key challenges remain in generalizability, scalability, and balancing fairness with overall model

performance.

Our project builds on these developments by empirically evaluating multiple approximate unlearning methods—each designed to modify only a small subset of parameters—across diverse datasets and bias dimensions. This comparative study is motivated by the need to understand the trade-offs involved in deploying unlearning strategies on small, resource-efficient models.

1. **Partitioned Contrastive Gradient Unlearning (PCGU):** PCGU optimizes only weight partitions most contributing to a specific bias domain by computing a first-order approximation on contrastive sentence pair gradients, focusing debiasing updates on targeted parameters. It was evaluated on gender–profession bias using association tests with StereoSet and CrowS metrics. PCGU is limited to masked language models and a single bias domain. It may also degrade language modeling ability outside its target area.
2. **Negative Preference Optimization (NPO):** NPO recasts unlearning as a negative-only preference optimization problem, employing a loss that penalizes high model probabilities on undesirable data to avoid the catastrophic utility collapse seen with gradient ascent. Experiments on synthetic datasets and the TOFU benchmark (forgetting 1–50% of synthetic author biographies) show improved utility preservation while effectively removing target data. Its reliance on synthetic benchmarks and removal-style tasks limits insights into real-world social bias mitigation
3. **Selective Knowledge negation Unlearning (SKU):** SKU is a two-stage framework that first identifies and acquires harmful knowledge, then negates it via targeted parameter updates while preserving performance on normal prompts. It was tested across various LLM architectures using harmful versus non-harmful prompt sets to assess dangerous content removal without utility loss. SKU’s effectiveness depends on accurate harmful knowledge identification and lacks evaluation on standard bias benchmarks, constraining its demonstrated fairness benefits
4. **Low Rank Adaptation Unlearning(LoRA):** LoRA injects trainable low-rank matrices into

the model’s attention layers, freezing the original weights and updating only these adapters to steer outputs away from harmful content. It was evaluated on bias and toxicity benchmarks like CrowS Pairs, StereoSet, and Real-ToxicityPrompts, showing significant reductions in bias scores and toxicity with negligible change in perplexity. By confining updates to small matrices, LoRA enables efficient, reversible unlearning, but may underperform on biases encoded in complex, high-rank parameter interactions.

2 Dataset and Methodology

2.1 Datasets used

To support our bias mitigation experiments, we selected and finalized multiple datasets that serve distinct roles in training and evaluation pipelines:

- **Bias Benchmark for QA (BBQ) (Parrish et al., 2022):** Includes 10,000 question-answer pairs annotated for stereotypes across nine social dimensions such as race, gender, and religion. We used it both for adversarial training via label inversion and for post-unlearning evaluation of stereotype likelihood.
- **CrowS-Pairs (Nangia et al., 2020):** Contains 1,508 sentence pairs contrasting stereotypical and counter-stereotypical constructions. We used it to evaluate bias propagation by comparing perplexity scores across paired completions.
- **PKU-SafeRLHF (Ji et al., 2024):** Provides over 44,600 prompts and 265,000 completions labeled across 19 harm categories. Includes dual-preference labels that decouple helpfulness from harmlessness for safety-aware evaluation of model outputs.
- **LLM LAT Harmful Dataset (Sheshadri et al., 2024):** Contains 5,000 prompts along with harmful and harmless completions, generated using the Self-Instruct framework (Wang et al., 2023). This dataset was used to assess the model’s reduction in harmful output generation post-unlearning.
- **TOFU Dataset (Maini et al., 2024):** Comprises question-answer pairs built from synthetic autobiographies generated by GPT-4. Used for evaluating selective knowledge unlearning in a controlled, non-sensitive setting.

- **WinoGender** (Rudinger et al., 2018): A diagnostic dataset designed to measure gender bias in pronoun resolution tasks. Used to observe shifts in gendered preference pre- and post-unlearning.
- **StereoSet** (Nadeem et al., 2021): Contains sentence-level completions assessing stereotypical versus anti-stereotypical preferences across race, gender, profession, and religion. Used to evaluate stereotype reinforcement tendencies in language generation.

2.2 Methodology & Approach

We explored multiple unlearning strategies on three compact architectures-Phi 3.5 (2.8 B), Qwen2.5 (0.5 B), and Mamba State-Space (130 M)-to accelerate iteration and reduce computational cost. All methods share this backbone selection, enabling apples-to-apples comparison of bias-removal efficacy versus compute expenditure.

2.2.1 Anisotropic Task Vector Negation (ATVN)

Low-Rank Adaptation (LoRA) introduces trainable low-rank matrices into the model’s architecture to facilitate efficient fine-tuning. Each LoRA module comprises two matrices: $\mathbf{A} \in R^{d \times r}$ and $\mathbf{B} \in R^{r \times d}$, where $r \ll d$. The product $\mathbf{W}_{\text{LoRA}} = \mathbf{AB}$ represents the task-specific adaptation.

To analyze and manipulate these adaptations, we perform Singular Value Decomposition (SVD) on $\mathbf{B}$:

$$\mathbf{B} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top \quad (1)$$

Here, $\mathbf{U} \in R^{r \times r}$ and $\mathbf{V} \in R^{d \times r}$ are orthogonal matrices, and $\mathbf{\Sigma} \in R^{r \times r}$ is a diagonal matrix containing the singular values.

By reconstructing $\mathbf{B}$ using only the top- r singular values and corresponding vectors, we obtain a low-rank approximation:

$$\mathbf{B}_r = \mathbf{U}_r \mathbf{\Sigma}_r \mathbf{V}_r^\top \quad (2)$$

This decomposition isolates the principal components of the adaptation, facilitating targeted manipulation.

- **Anisotropic Scaling for Task Vector Negation**

Recognizing that different components of the model (e.g., attention layers, feed-forward

networks, embedding layers) contribute variably to task-specific adaptations, we introduce anisotropic scaling factors to modulate the negation process selectively.

Let λ_{attn} , λ_{mlp} , and λ_{embed} denote the scaling factors for attention, MLP, and embedding layers, respectively. Additionally, let λ_{bias} represent the scaling factor applied when a bias term is present. The negated adaptation for a given layer is computed as:

$$\mathbf{W}_{\text{neg}} = -\lambda \cdot \mathbf{U}_r \mathbf{\Sigma}_r \mathbf{V}_r^\top \quad (3)$$

where λ is chosen based on the layer type:

$$\lambda = \begin{cases} \lambda_{\text{attn}}, & \text{if layer is attention} \\ \lambda_{\text{mlp}}, & \text{if layer is MLP} \\ \lambda_{\text{embed}}, & \text{if layer is embedding} \\ \lambda_{\text{bias}}, & \text{if bias term is present} \end{cases}$$

This targeted scaling ensures that each component’s contribution to the overall adaptation is appropriately adjusted, allowing for more precise unlearning.

- **Adaptive Hyperparameter Optimization via HyperQ-Opt**

Selecting optimal scaling factors λ is critical for effective task vector negation. To automate this selection, we employ an adaptive hyperparameter optimization strategy inspired by Q-learning, termed *HyperQ-Opt*.

Q-Learning Framework

We define a discrete action space $\mathcal{A}$ comprising candidate scaling factor tuples $(\lambda_{\text{weight}}, \lambda_{\text{bias}})$. The Q-value for each action $a \in \mathcal{A}$ is initialized to zero:

$$Q(a) = 0, \quad \forall a \in \mathcal{A} \quad (4)$$

At each iteration, an action a_t is selected using an ϵ -greedy policy:

$$a_t = \begin{cases} \text{random action from } \mathcal{A}, & \text{with probability } \epsilon \\ \arg \max_{a \in \mathcal{A}} Q(a), & \text{with probability } 1 - \epsilon \end{cases}$$

After applying the selected scaling factors and evaluating the resulting model, we obtain a

reward r_t reflecting the effectiveness of the negation. The Q-value is then updated as:

$$Q(a_t) \leftarrow Q(a_t) + \alpha (r_t - Q(a_t)) \quad (5)$$

where α is the learning rate.

- **Reward Function**

The reward function is designed to quantify the improvement in bias mitigation achieved by the negation. Let S_{base} and S_{neg} denote the bias scores before and after negation, respectively. The reward is computed as:

$$r_t = \sum_i w_i (|S_{\text{base},i}| - |S_{\text{neg},i}|) \quad (6)$$

where w_i is a weight reflecting the importance of the i -th bias category.

- **Convergence Criteria**

The optimization process continues until convergence, determined by the standard deviation of recent rewards falling below a predefined threshold δ :

$$\text{Converged if } \text{std}(r_{t-k+1}, \dots, r_t) < \delta$$

where k is the number of recent iterations considered.

Conclusion

The proposed methodology integrates SVD-based decomposition of LoRA adaptations, anisotropic scaling tailored to model components, and adaptive hyperparameter optimization via HyperQ-Opt. This comprehensive approach facilitates precise and efficient task vector negation, enabling the unlearning of specific behaviors or knowledge in SLMs without extensive retraining.

2.2.2 Partitioned Contrastive Gradient Unlearning (PCGU)

Partitioned Contrastive Gradient Unlearning (PCGU) is an extension of contrastive learning and gradient-based optimization techniques, designed to remove specific learned biases from pretrained language models (PLMs) such as Mamba 0.13B (LoRA). The goal of PCGU is to unlearn biased patterns while maintaining the model’s overall performance and utility on other tasks.

The key steps involved in PCGU are:

- **Contrastive Loss:** We use a contrastive loss function that compares biased and less-biased model outputs. This loss is designed to encourage the model to generate less biased outputs post-unlearning. Given two sentences, one with a stereotype and the other with an anti-stereotype, the loss is computed as:

$$\mathcal{L}_{\text{contrastive}} = \max(0, \text{loss}_{\text{stereotype}} - \text{loss}_{\text{anti-stereotype}}) \quad (7)$$

This ensures that the model prioritizes anti-stereotypical completions over stereotypical ones.

- **Regularization:** To prevent the unlearning process from distorting the model’s general behavior, we incorporate a regularization term to the loss function. The regularization term is computed based on the difference between the model’s weights before and after the unlearning process:

$$\mathcal{L}_{\text{reg}} = \sum_i \|W_{\text{unlearned}} - W_{\text{fine-tuned}}\|^2 \quad (8)$$

This prevents overfitting to biased data and ensures that the model retains useful knowledge while eliminating harmful biases.

- **Gradient Updates:** Once the contrastive loss and regularization term are calculated, we update the model’s parameters using the AdamW optimizer. To improve training efficiency, mixed-precision training is utilized. Additionally, gradient clipping is applied to prevent gradient explosion, ensuring stable training and effective bias reduction.
- **Projection Steps:** We run the optimization process for 30 projection steps. These steps guide the model towards unlearning biased associations while ensuring that general knowledge is retained, preventing complete unlearning or overcorrection.

Conclusion

The proposed methodology integrates **contrastive loss**, **regularization**, and **gradient updates** to perform PCGU, effectively mitigating biased behavior in pretrained language models. This approach allows for the targeted unlearning of harmful knowledge without requiring extensive retraining. By combining contrastive learning with gradient-based optimization, PCGU provides a computationally

efficient and precise method for debiasing language models. This methodology proves that **Partitioned Contrastive Gradient Unlearning (PCGU)** is a powerful tool for debiasing, with the results indicating significant improvements in bias mitigation on datasets like **StereoSet**. The approach not only reduces biases but also maintains the model’s fluency and overall general utility.

3 Experiments

3.1 Anisotropic Task Vector Negation

We evaluated the efficacy of task vector negation in mitigating social biases using two pre-trained models: Phi-3.5 (2.8B parameters) and Qwen2.5 (0.5B parameters). Both models were fine-tuned using Low-Rank Adaptation (LoRA) (Hu et al., 2021), which introduces trainable low-rank matrices into Transformer layers, enabling efficient adaptation with reduced computational overhead.

The Bias Benchmark for Question Answering (BBQ) dataset (Parrish et al., 2022) was employed to assess and mitigate social biases. BBQ comprises question sets targeting nine social dimensions relevant to U.S. English-speaking contexts, designed to expose model biases.

3.1.1 Experimental Setup

- **Model Preparation:** Models were quantized to 4-bit precision to optimize memory usage and computational efficiency. Specific modules were targeted for LoRA adaptation.
- **LoRA Configuration:** For Phi-3.5, LoRA targeted the qkv_{proj} and o_{proj} modules with rank $r = 8$ and scaling factor $\alpha = 32$. A dropout rate of 0.1 was applied. Qwen2.5 employed a similar configuration, adjusted as per its architecture.
- **Data Processing:** Training and evaluation datasets were derived from BBQ, formatted into input-output pairs suitable for causal language modeling. Tokenization included truncation and padding to a maximum sequence length of 512 tokens.
- **Training Procedure:** Models were fine-tuned over three epochs with a learning rate of 2×10^{-4} and batch size of 8, utilizing mixed precision (FP16) training.

3.1.2 Evaluation Metrics

- **Bias Score:** Quantifies the degree of bias in model predictions, calculated based on the disparity in responses across different social groups within BBQ.
- **Accuracy:** Measures the correctness of model responses to BBQ questions, indicating any trade-offs between bias mitigation and performance.

3.1.3 Experimental Procedure

1. **Baseline Evaluation:** Assessed initial bias and accuracy of pre-trained models on BBQ without fine-tuning.
2. **LoRA Fine-Tuning:** Fine-tuned models on BBQ using LoRA to adapt to the task and potentially reduce inherent biases.
3. **Task Vector Negation:** Computed the difference between fine-tuned and pre-trained model parameters to isolate task-specific bias vectors, which were then subtracted to mitigate learned biases.
4. **Post-Negation Evaluation:** Evaluated bias score and accuracy post-negation to determine the effectiveness of bias mitigation.

3.2 Partitioned Contrastive Gradient Unlearning (PCGU)

To investigate the effectiveness of **Partitioned Contrastive Gradient Unlearning (PCGU)** in mitigating biases, we conducted experiments using a pretrained model, Mamba 0.13B (LoRA). The model was fine-tuned using the LoRA technique (Hu et al., 2021) for efficient adaptation. The **WinoGender** dataset (Rudinger et al., 2018), which focuses on gender bias in pronoun resolution tasks, was used for fine-tuning. After fine-tuning, the model’s performance and bias reduction capabilities were evaluated using the **StereoSet** dataset (Nadeem et al., 2021), which contains biased and anti-biased pairs for gender, race, and occupation stereotypes. We leveraged the StereoSet dataset to assess the model’s ability to unlearn biased associations while maintaining its performance across other tasks.

The fine-tuning and unlearning process involved the following steps:

1. **Model Preparation:** The Mamba 0.13B model was loaded and optimized for fine-tuning. Mixed-precision training was applied to enhance efficiency and computational performance.
2. **LoRA Fine-Tuning:** The model was fine-tuned using the LoRA technique on the WinoGender dataset, focusing on reducing inherent biases across gender pronouns.
3. **PCGU Unlearning:** PCGU was applied to the fine-tuned model using contrastive loss, regularization, and gradient updates to mitigate bias. The optimization process was run for 30 projection steps to gradually unlearn biased associations while retaining useful knowledge.
4. **Post-Unlearning Evaluation:** After the PCGU unlearning process, the model was evaluated using the StereoSet dataset to assess improvements in bias mitigation.
2. **LoRA Fine-Tuning:** Fine-tuned the model on the WinoGender dataset to adjust for gender-based bias in pronoun resolution tasks and potentially reduce inherent biases.
3. **PCGU Unlearning:** Applied PCGU by updating the model’s parameters using contrastive loss, regularization, and gradient updates to remove the biased patterns learned during fine-tuning.
4. **Post-Unlearning Evaluation:** Evaluated the bias score and qualitative prompt-based evaluation on the debiased model to measure improvements in bias mitigation.

4 Results and Analysis

5 Shortcomings and Future Work

Limitations

Model Weaknesses

Certain techniques that eliminate bias from the model simultaneously reduce the model’s confidence and fluency in its responses. For instance, models frequently provide less clear or better responses following unlearning.

Furthermore, not all forms of prejudice respond as well to these techniques. They do better when it comes to gender and race bias, but less well when it comes to sexual orientation or religion.

Dataset Problems

A number of datasets, such as TOFU and LLM LAT, are artificial and do not accurately represent the actual world. This implies that results may not translate well to real-world application scenarios.

Additionally, some groups, such as elderly folks or those with impairments, lack sufficient examples, making it more difficult to assess justice for them.

Future Work

Smarter Unlearning Methods

We plan to improve how the model decides how much to unlearn and from where. Smarter methods like *HyperQ-Opt*, which uses reinforcement learning, can help make unlearning more targeted and effective.

The goal is to reduce bias without making the model worse at language tasks.

3.2.1 Evaluation Metrics

To evaluate the impact of PCGU on model bias and performance, we employed the following evaluation metrics:

- **StereoSet Bias Score:** Measures the degree of bias in the model’s output in terms of stereotype, anti-stereotype, and unrelated responses. The bias score was computed before and after unlearning to assess the improvement in reducing biased outputs.
- **Qualitative Prompt-Based Evaluation:** In addition to the StereoSet bias score, we evaluated the model on a set of qualitative prompts related to gender, race, and occupation stereotypes. This evaluation assesses the model’s ability to produce non-biased completions compared to pre-defined unbiased expectations.

3.2.2 Experiments Conducted

We performed the following experiments to assess the effectiveness of PCGU:

1. **Baseline Evaluation:** We assessed the initial bias and general performance of the pretrained Mamba 0.13B (LoRA) model on the StereoSet dataset after fine-tuning on the WinoGender dataset, but before applying any PCGU unlearning.

Better Evaluation Tools

We'll continue building *UNLEARN-BENCH*, a tool to test and compare unlearning methods in a fair and complete way.

This includes adding tests for different languages, cultures, and real-world scenarios, as well as checking how models respond to tricky or biased prompts.

6 Artifacts

References

- Edward J. Hu et al. 2021. [Lora: Low-rank adaptation of large language models](#). *arXiv preprint arXiv:2106.09685*.
- Jiaming Ji, Donghai Hong, Borong Zhang, Boyuan Chen, Josef Dai, Boren Zheng, Tianyi Qiu, Boxun Li, and Yaodong Yang. 2024. [Pku-saferlhf: Towards multi-level safety alignment for llms with human preference](#). *Preprint*, arXiv:2406.15513.
- Pratyush Maini, Zhili Feng, Avi Schwarzschild, Zachary C. Lipton, and J. Zico Kolter. 2024. Tofu: A task of fictitious unlearning for llms.
- Moin Nadeem, Anna Bethke, and Siva Reddy. 2021. [Stereoset: Measuring stereotypical bias in pretrained language models](#). In *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics (ACL)*, pages 5356–5371. Association for Computational Linguistics.
- Nikita Nangia et al. 2020. [Crows-pairs: A challenge dataset for measuring social biases in masked language models](#). *arXiv preprint arXiv:2010.00133*.
- Alicia Parrish, Angelica Chen, Nikita Nangia, Vishakh Padmakumar, Jason Phang, Jana Thompson, Phu Mon Htut, and Samuel R. Bowman. 2022. [Bbq: A hand-built bias benchmark for question answering](#). *Preprint*, arXiv:2110.08193.
- Rachel Rudinger, Jason Naradowsky, Brian Leonard, and Benjamin Van Durme. 2018. [Gender bias in coreference resolution](#). In *Proceedings of the 2018 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 2 (Short Papers)*, pages 8–14. Association for Computational Linguistics.
- Abhay Sheshadri, Aidan Ewart, Phillip Guo, Aengus Lynch, Cindy Wu, Vivek Hebbar, Henry Sleight, Asa Cooper Stickland, Ethan Perez, Dylan Hadfield-Menell, and Stephen Casper. 2024. Targeted latent adversarial training improves robustness to persistent harmful behaviors in llms. *arXiv preprint arXiv:2407.15549*.
- Yizhong Wang, Yeganeh Kordi, Swaroop Mishra, Alisa Liu, Noah A. Smith, Daniel Khashabi, and Hannaneh

Hajishirzi. 2023. [Self-instruct: Aligning language models with self-generated instructions](#). *Preprint*, arXiv:2212.10560.